The Illusion of Digital Consent

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Research Question

- While smartphone users “consent” to location data collection, do they actually understand what they’re agreeing to?

- Does the structure of digital consent give them meaningful control?





01

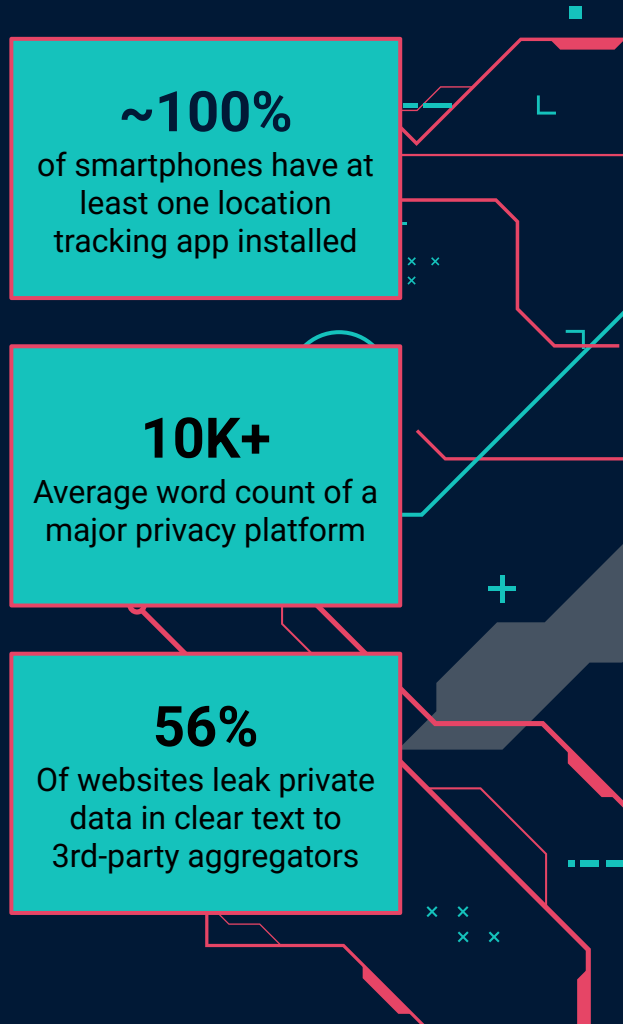
Introduction

What you’re actually agreeing to…

Introduction

- Phones collect geolocation data continuously through private policies that users agree and don't agree to
- What does consent mean when policies are 10k+ words, or when the only options are "agree" or "leave"?
- The structure of digital consent + data economies is limited, uneven, even an "illusion"

(Custers et al.; Chen & Chen, 2025; Jonker & Brits, 2025; Tran et al., 2012.)





02

The Issues

How they’re doing it...

How You're Getting Tracked

- GPS and Cell Towers
  - Pinpoints location 24/7
- Wifi triangulation
  - Maps indoor position
- Third-Party Scripts
  - JS trackers on 1M+ sites collect data covertly

- Cookie Banners
  - Tracks before given consent and after being denied
- App Permissions
  - Broad permissions often get bundled
- Data Brokers
  - Location data sold + aggregated far beyond its initial purpose

(Knox, 2019; Tran et al. 2012; Paci et al.; Chen & Chen, 2025; Custers et al.)

Why Informed Consent Is Broken

- Privacy Fatigue
  - Users usually click ‘agree’ out of exhaustion, not understanding
- The Illusion of Choice
  - Most platforms have the ultimatum of agreeing to their policies or not using their services
- The Privacy Paradox
  - Users claim they care about privacy but share their info anyway

“Privacy fatigue increases inconsistency among cognition, affect, and behavior, undermining consent effectiveness.” -Chen & Chen, 2025

(Chen & Chen, 2025; Custers et al.; Jonker & Brits, 2025)



Legal Gaps & Structural Power Imbalance

Carpenter v. United States (2018)

- Supreme Court ruled that accessing cell site location data without a warrant violates 4th Amendment
- Knox (2019) argues this ruling signals broader protections for data held by 3rd parties

Key Implication: Even ‘voluntary’ data sharing may still carry a reasonable expectation of privacy

GDPR vs. United States’ Patchwork

EU (GDPR):

- Explicit opt-in consent required before any tracking

US Law:

- Patchwork of state laws, mostly opt-out mechanisms

Both:

- No examined sites from either fully complied with GDPR consent standards

(Knox, 2019; Paci et al.)



03

The Trade-Off

Of Convenience vs. Control...

Trade-Off:

Benefits ✔

- Navigation and real-time traffic (Google/Apple Maps, Waze, etc.)
- Emergency services/911 location accuracy
- Contact tracing and public health monitoring
- Personalized services valued by users
- IoT and smart city infrastructure efficiency

Limitations ❌

- Consent UI designed to maximize data collection, not understanding
- Privacy tools remain inaccessible to most users
- Data repurposed without “re-consent”
- Location data has been used for policing and targeting marginalized groups/races
- Choosing no can take away from the app’s core functionality

(Jonker & Brits, 2025; Emami-Naeini et al., 2018; Chen & Chen, 2025; Cranor et al.; Custers et al.; Knox, 2019; Paci et al.)



04

My Analysis

From an STS Perspective...

My Analysis

Technology isn't Neutral

- Location tracking systems are designed within political economies
- Consent UI, data retention policies, and permission architectures are designed by corps to maximize data collection

Unequal Surveillance

- Location data disproportionately used to surveil minority communities, low-income populations, and immigrants
- 4th Amendment after Carpenter are still unevenly applied
- Marginalized groups bear higher cost from data economy

Data Economy as Infrastructure

- Data sharing almost not optional for digital participation
- IoT users get more heavily influenced by social norms
- Privacy decisions happen in systems that constrain options

(Chen & Chen, 2025; Paci et al.; Knox, 2019; Emami-Naeini et al., 2018; Jonker & Brits, 2025)



05

Conclusion

Consensus and Recommendations...

Conclusion:

Smartphone users appear to have control over their data, but the structure of digital consent and data economies makes that control limited, uneven, and manufactured

Recommendations:

Reform Consent Architecture	Extend Legal Protections	Center Equity
- Redesign privacy pop-ups for understanding, not compliance	- Further enforce the Carpenter v. United States rulings - Location data needs more federal protection	- Ensure privacy reform accounts for how disproportionate data tracking has burdened marginalized communities

(Chen & Chen, 2025; Cranor et al.; Knox, 2019; Paci et al.; Emami-Naeimi et al.)



THANKS!

Do you have any questions?



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